

## 19. Rectifiability and analytic capacity in the complex plane

### Analytic capacity and removable sets

In this chapter we shall discuss a classical problem in complex analysis and its relations to the rectifiability of sets in the complex plane  $\mathbf{C}$ . The problem is the following: which compact sets  $E \subset \mathbf{C}$  are removable for bounded analytic functions in the following sense?

(19.1) If  $U$  is an open set in  $\mathbf{C}$  containing  $E$  and  $f: U \setminus E \rightarrow \mathbf{C}$  is a bounded analytic function, then  $f$  has an analytic extension to  $U$ .

This problem has been studied for almost a century, but a geometric characterization of such removable sets is still lacking. We shall prove some partial results and discuss some other results and conjectures. For many different function classes a complete solution has been given in terms of Hausdorff measures or capacities. For example, if the boundedness is replaced by the Hölder continuity with exponent  $\alpha$ ,  $0 < \alpha \leq 1$ , then the necessary and sufficient condition for the removability of  $E$  is that  $\mathcal{H}^{1+\alpha}(E) = 0$ , see Exercise 4, Dolzenko [1] and Uy [2], and for the corresponding question for harmonic functions Carleson [1]. Král [1] proved that for the analytic BMO functions the removable sets  $E$  are characterized by the condition  $\mathcal{H}^1(E) = 0$ . The problem (19.1) is more delicate, because the metric size is not the only thing that matters; the rectifiability structure also seems to be essential as we shall see.

Ahlfors [1] introduced a set function  $\gamma$ , called *analytic capacity*, whose null-sets are exactly the removable sets of (19.1). It is defined for compact sets  $E \subset \mathbf{C}$  by

(19.2)  $\gamma(E) = \sup \{ |f'(\infty)| : f \text{ is analytic in } \mathbf{C} \setminus E \text{ with } \|f\|_\infty \leq 1 \}.$

Here

$$\|f\|_\infty = \sup \{ |f(z)| : z \in \mathbf{C} \setminus E \}$$

and

$$f'(\infty) = \lim_{z \rightarrow \infty} z(f(z) - f(\infty))$$

with

$$f(\infty) = \lim_{z \rightarrow \infty} f(z).$$

(Usually  $f'(\infty) \neq \lim_{z \rightarrow \infty} f'(z)$ .)

Note that if  $f$  is as in (19.2) and

$$g(z) = \frac{f(z) - f(\infty)}{1 - \overline{f(\infty)}f(z)},$$

then  $\|g\|_\infty \leq 1$ ,  $g(\infty) = 0$  and  $g'(\infty) = f'(\infty)/(1 - |f(\infty)|^2)$ . Thus in the definition of  $\gamma(E)$  one can restrict to functions vanishing at infinity.

**19.3. Theorem.** *For a compact set  $E \subset \mathbf{C}$  the following conditions are equivalent:*

- (1)  $\gamma(E) = 0$ .
- (2) Every bounded analytic function  $f: \mathbf{C} \setminus E \rightarrow \mathbf{C}$  is constant.
- (3) The condition (19.1) holds.

Moreover, these conditions imply that  $E$  is totally disconnected.

*Proof.* We first verify that  $E$  is totally disconnected if (2) holds. In fact, if  $E$  should contain a non-degenerate continuum  $K$ , we could use the Riemann mapping theorem to map each component of  $\mathbf{C} \setminus K$  conformally onto  $U(1)$ , which would give a non-constant bounded analytic function in  $\mathbf{C} \setminus K$  violating (2). Clearly (2) implies (1). On the other hand if (2) fails, there is a bounded analytic function  $f: \mathbf{C} \setminus E \rightarrow \mathbf{C}$  with  $f(\infty) = 0$  and  $f(z_0) \neq 0$  for some  $z_0 \in \mathbf{C} \setminus E$ . Defining

$$g(z) = \frac{f(z) - f(z_0)}{z - z_0} \quad \text{for } z \in \mathbf{C} \setminus E, \ z \neq z_0,$$

and  $g(z_0) = f'(z_0)$ , we see that  $g$  is bounded and analytic in  $\mathbf{C} \setminus E$  with  $g'(\infty) = -f(z_0)$ . Thus  $\gamma(E) > 0$ . So (1) and (2) are equivalent.

To see that (2) implies (3), let  $U$  be an open set containing  $E$  and let  $f$  be bounded and analytic in  $U \setminus E$ . Let  $z \in U \setminus E$ . As  $E$  is totally disconnected, we can choose smooth disjoint Jordan curves  $\Gamma_1$  and  $\Gamma_2$  in  $U$  both surrounding  $E$  such that  $z$  lies inside  $\Gamma_1$  and outside  $\Gamma_2$ . By the Cauchy integral formula,

$$f(z) = \frac{1}{2\pi i} \int_{\Gamma_1} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \int_{\Gamma_2} \frac{f(\zeta)}{\zeta - z} d\zeta = f_1(z) + f_2(z).$$

As long as the above properties are in force,  $f_1(z)$  and  $f_2(z)$  are independent of  $\Gamma_1$  and  $\Gamma_2$ . Moreover, we may use these formulas to define  $f_1(z)$  for all  $z \in U$  and  $f_2(z)$  for all  $z \in \mathbf{C} \setminus E$ . Then  $f_1$  is analytic in  $U$  and  $f_2$  bounded and analytic in  $\mathbf{C} \setminus E$ . By (2),  $f_2 \equiv f_2(\infty) = 0$ , whence  $f_1$  is an analytic extension of  $f$  to  $U$ .

Finally, if (3) holds it is an easy exercise to show that  $E$  cannot have interior points. Thus the extended functions are also bounded. Consequently, every bounded analytic function in  $\mathbf{C} \setminus E$  extends to a bounded analytic function in  $\mathbf{C}$ , which by Liouville's theorem must be constant.  $\square$

Although analytic capacity gives a solution to the problem (19.1), this solution is not geometric. Analytic capacity is an entirely complex-analytic concept and there are many unsolved problems about it. For example, it is not known whether it is subadditive or whether its null-sets are invariant under affine mappings. However, it has turned out to be very useful in the theory of rational approximation, see Garnett [2], Zalcman [1], Verdera [3] and Vitushkin [2].

### Analytic capacity, Riesz capacity and Hausdorff measures

We shall now give two simple relations between analytic capacity and Hausdorff measures. Recall that  $\dim E > 1$  implies  $C_1(E) > 0$  for Borel sets by Theorem 8.9.

**19.4. Theorem.** *If  $E \subset \mathbf{C}$  is compact and  $C_1(E) > 0$ , then  $\gamma(E) > 0$ .*

*Proof.* Since  $C_1(E) > 0$  there is a Radon measure  $\mu$  with  $\text{spt } \mu \subset E$ ,  $0 < \mu(E) < \infty$ , and

$$\int \frac{d\mu\zeta}{|\zeta - z|} \leq 1 \quad \text{for } z \in \mathbf{C}.$$

(Take a suitable restriction of a measure  $\nu$  with  $I_1(\nu) < \infty$  (see Exercise 8.2).) Setting

$$f(z) = \int \frac{d\mu\zeta}{\zeta - z}, \quad z \in \mathbf{C} \setminus E,$$

a direct computation shows that  $f$  has complex derivative in  $\mathbf{C} \setminus E$ , whence it is analytic,  $f(\infty) = 0$ , and

$$f'(\infty) = \lim_{z \rightarrow \infty} \int \frac{1}{\zeta/z - 1} d\mu\zeta = -\mu(E) \neq 0.$$

Thus  $\gamma(E) > 0$ .  $\square$

In the other direction we have a theorem of Painlevé from the last century.

**19.5. Theorem.** *If  $E \subset \mathbf{C}$  is compact and  $\mathcal{H}^1(E) = 0$ , then  $\gamma(E) = 0$ .*

*Proof.* Let  $z \in \mathbf{C} \setminus E$  and let  $0 < \varepsilon < d(z, E)/2$ . We can cover the compact set  $E$  with discs  $B_j$ ,  $j = 1, \dots, k$ , such that  $E \cap B_j \neq \emptyset$  and  $\sum_{j=1}^k d(B_j) < \varepsilon$ . Let  $f: \mathbf{C} \setminus E \rightarrow \mathbf{C}$  be analytic with  $\|f\|_\infty \leq 1$  and  $f(\infty) = 0$ . Choosing  $R$  such that  $E \cup \{z\} \subset B(R)$  and letting  $\Gamma = \partial(\bigcup_{j=1}^k B_j)$ , we have by the Cauchy integral formula

$$f(z) = \frac{1}{2\pi i} \int_{S(R)} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Since  $f(\infty) = 0$ , the first integral tends to zero as  $R$  tends to infinity. Thus

$$|f(z)| \leq \frac{1}{2\pi} \int_{\Gamma} \frac{|f(\zeta)|}{|\zeta - z|} d\mathcal{H}^1\zeta \leq \frac{1}{\pi d(z, E)} \sum_{j=1}^k \mathcal{H}^1(\partial B_j) \leq \frac{\varepsilon}{d(z, E)}.$$

Letting  $\varepsilon \downarrow 0$ , we obtain  $f(z) = 0$ . Hence  $\gamma(E) = 0$ . □

Our main interest here is to know which compact sets have zero analytic capacity. The two simple theorems above show that we only have to worry about sets of Hausdorff dimension one. We shall mainly pay attention to sets of finite  $\mathcal{H}^1$  measure and later on give some comments on those having infinite, or more essentially, non- $\sigma$ -finite,  $\mathcal{H}^1$  measure.

We shall see that there are many sets of positive  $\mathcal{H}^1$  measure and zero analytic capacity. But before this we briefly look at an important class of sets for which  $\mathcal{H}^1$  and  $\gamma$  are simultaneously zero. We give the proof for the following deep theorem only in a very special case.

**19.6. Theorem.** *Let  $\Gamma \subset \mathbf{C}$  be a rectifiable curve and  $E$  a compact subset of  $\Gamma$ . Then  $\gamma(E) = 0$  if and only if  $\mathcal{H}^1(E) = 0$ .*

*Proof for  $\Gamma \subset \mathbf{R}$ .* Let  $E \subset \mathbf{R} \subset \mathbf{C}$  with  $\mathcal{L}^1(E) > 0$ . Set

$$g(z) = \int_E \frac{1}{x - z} d\mathcal{L}^1x, \quad z \in \mathbf{C} \setminus E.$$

By a direct computation the values of  $g$  are contained in the strip  $S = \{x + iy : |y| < \pi\}$ . Let  $h: S \rightarrow U(1)$  be a conformal map. Then  $f = h \circ g$  is a bounded non-constant analytic function in  $\mathbf{C} \setminus E$ , whence  $\gamma(E) > 0$ . □

**19.7. Remark.** The above proof gives an estimate  $\gamma(E) \geq c\mathcal{L}^1(E)$  for  $E \subset \mathbf{R}$ . One can show more precisely that  $\gamma(E) = \mathcal{L}^1(E)/4$  for compact subsets  $E$  of  $\mathbf{R}$ , see Garnett [2, I.6.2].

In the general case the proof for Theorem 19.6 has been given in several stages. See Christ [1], Marshall [1] and Murai [2] for various aspects of this. Relying on the earlier works of Havin and Havinson [1], Havin [1] and Davie [1], the final step needed to complete the proof was the theorem of Calderón [1] stating that the Cauchy transform,  $f \mapsto \int_{\Gamma} (\zeta - z)^{-1} f(\zeta) d\zeta$ , defines a bounded operator in  $L^2$  on Lipschitz graphs with small Lipschitz constant, see Remark 19.18 (5), Theorem 20.15 and the discussion following it.

An immediate consequence of Theorem 19.6 is that if  $E$  is a compact subset of  $\mathbf{C}$  with  $\mathcal{H}^1(E) < \infty$  and  $\gamma(E) = 0$ , then  $E$  is purely 1-unrectifiable. Indeed, otherwise  $\mathcal{H}^1(E \cap \Gamma) > 0$  for some rectifiable curve  $\Gamma$  and so  $\gamma(E) > 0$ . A reasonable conjecture seems to be that the converse also holds.

**19.8. Conjecture.** Let  $E \subset \mathbf{C}$  be compact with  $\mathcal{H}^1(E) < \infty$ . Then  $\gamma(E) = 0$  if and only if  $E$  is purely 1-unrectifiable.

## Cauchy transforms of complex measures

We now head towards a partial result on this (see Theorem 19.17 later). In this chapter we shall also use complex measures; all we need about them can be found in Rudin [1]. In particular,  $|\sigma|$  is the variation measure of the complex measure  $\sigma$  defined for Borel sets  $B \subset \mathbf{C}$  by

$$|\sigma|(B) = \sup \left\{ \sum_{i=1}^k |\sigma(B_i)| : B_1, \dots, B_k \text{ are disjoint Borel sets with } B = \bigcup_{i=1}^k B_i \right\},$$

and  $\text{spt } \sigma = \text{spt } |\sigma|$ . Note that  $|\sigma(B)| \leq |\sigma|(B)$  and the inequality may be strict.

**19.9. Theorem.** Let  $E \subset \mathbf{C}$  be compact with  $\mathcal{H}^1(E) < \infty$  and let  $f: \mathbf{C} \setminus E \rightarrow \mathbf{C}$  be analytic with  $\|f\|_{\infty} \leq 1$  and  $f(\infty) = 0$ . Then there is a complex Radon measure  $\sigma$  such that  $\text{spt } \sigma \subset E$ ,

$$|\sigma(B(z, r))| \leq r \quad \text{for } z \in \mathbf{C} \text{ and } r > 0$$

and

$$f(z) = \int \frac{d\sigma\zeta}{\zeta - z} \quad \text{for } z \in \mathbf{C} \setminus E.$$

Moreover, there is a Borel function  $\varphi: E \rightarrow \mathbf{C}$  such that  $|\varphi| \leq 1$ ,  $\sigma(A) = \int_A \varphi d\mathcal{H}^1$  for Borel sets  $A \subset E$  and

$$f(z) = \int_E \frac{\varphi(\zeta)}{\zeta - z} d\mathcal{H}^1\zeta \quad \text{for } z \in \mathbf{C} \setminus E.$$

*Proof.* Repeating the argument of the proof of Theorem 19.5 with  $\varepsilon = 1/k$ ,  $k = 1, 2, \dots$ , we can cover  $E$  with closed discs  $B_{k,1}, \dots, B_{k,m_k}$  such that  $d(B_{k,j}) < 1/k$ ,

$$\sum_{j=1}^{m_k} d(B_{k,j}) \leq 2\mathcal{H}^1(E) + 1/k,$$

and, setting  $\Gamma_k = \partial(\bigcup_{j=1}^{m_k} B_{k,j})$ ,

$$(1) \quad f(z) = -\frac{1}{2\pi i} \int_{\Gamma_k} \frac{f(\zeta)}{\zeta - z} d\zeta$$

for  $z \in \mathbf{C} \setminus \bigcup_{j=1}^{m_k} B_{k,j}$ . Consider the complex Radon measures  $\sigma_k$  determined by

$$\int \psi d\sigma_k = -\frac{1}{2\pi i} \int_{\Gamma_k} \psi(\zeta) f(\zeta) d\zeta$$

for continuous functions  $\psi$ . Then (1) takes the form

$$f(z) = \int \frac{1}{\zeta - z} d\sigma_k\zeta$$

and the  $\sigma_k$ 's have uniformly bounded total variations:

$$\|\sigma_k\| = |\sigma_k|(\mathbf{C}) \leq \frac{1}{2\pi} \int_{\Gamma_k} |f(\zeta)| d\mathcal{H}^1\zeta \leq \mathcal{H}^1(E) + 1.$$

Hence by the analogue of Theorem 1.23 for complex measures we can extract a sub-sequence converging weakly to a complex measure  $\sigma$ . Obviously,  $\text{spt } \sigma \subset E$  and

$$f(z) = \int \frac{d\sigma\zeta}{\zeta - z} \quad \text{for } z \in \mathbf{C} \setminus E.$$

Let  $D$  be an open disc with

$$(2) \quad \mathcal{H}^1(E \cap \partial D) = 0 \quad \text{and} \quad \int_{\partial D} \int |\zeta - z|^{-1} d|\sigma| \zeta d\mathcal{H}^1 z < \infty.$$

As

$$-\frac{1}{2\pi i} \int_{\partial D} \frac{dz}{\zeta - z} = 1 \quad \text{for } \zeta \in D \quad \text{and} \quad = 0 \quad \text{for } \zeta \notin \overline{D},$$

we obtain by Fubini's theorem

$$\begin{aligned} -\frac{1}{2\pi i} \int_{\partial D} f(z) dz &= -\frac{1}{2\pi i} \int_{\partial D} \int \frac{d\sigma \zeta}{\zeta - z} dz \\ &= \int \left( -\frac{1}{2\pi i} \int_{\partial D} \frac{dz}{\zeta - z} \right) d\sigma \zeta = \sigma(D), \end{aligned}$$

whence

$$(3) \quad |\sigma(D)| \leq \frac{1}{2\pi} \mathcal{H}^1(\partial D) = \frac{1}{2} d(D).$$

Since  $\int_{B(r)} |z|^{-1} d\mathcal{L}^2 z < \infty$  for all  $0 < r < \infty$ , Fubini's theorem yields that (2), and hence also (3), holds for almost all discs with a fixed centre ( $\mathcal{H}^1(E \cap \partial D) > 0$  can happen for at most countably many radii). From this (3) follows for all discs  $D$  by approximation. Finally, (3) implies that

$$(4) \quad |\sigma|(A) \leq \mathcal{H}^1(A) \quad \text{for Borel sets } A \subset E.$$

To see this, let  $\varepsilon > 0$  and apply the density theorem 6.2 (1) and Vitali's covering theorem 2.8 to find disjoint discs  $B_i = B(x_i, r_i)$  such that

$$(1 + \varepsilon) \mathcal{H}^1(A \cap B_i) \geq r_i, \quad |\sigma|\left(A \setminus \bigcup_i B_i\right) = 0 \quad \text{and} \quad |\sigma|\left(\bigcup_i B_i \setminus A\right) < \varepsilon.$$

Then by (3),

$$\begin{aligned} |\sigma(A)| &\leq \left| \sigma\left(\bigcup_i B_i\right) \right| + \left| \sigma\left(\bigcup_i B_i \setminus A\right) \right| \leq \sum_i |\sigma(B_i)| + \varepsilon \\ &\leq \sum_i r_i + \varepsilon \leq (1 + \varepsilon) \sum_i \mathcal{H}^1(A \cap B_i) + \varepsilon \\ &\leq (1 + \varepsilon) \mathcal{H}^1(A) + \varepsilon. \end{aligned}$$

Thus  $|\sigma(A)| \leq \mathcal{H}^1(A)$  for all Borel sets  $A \subset E$  and (4) follows by the definition of the total variation.

From (4) we see that  $\sigma$  is absolutely continuous with respect to  $\mathcal{H}^1 \llcorner E$  with Radon-Nikodym derivative  $\varphi$  such that  $|\varphi| \leq 1$ . This proves the last statement.  $\square$

**19.10. Corollary.**  $\gamma(E) \leq \mathcal{H}^1(E)$ .

*Proof.* Since  $f'(\infty) = -\int_E \varphi d\mathcal{H}^1$ , this follows by the definition of  $\gamma$ .  $\square$

**19.11. Remarks.** (1) Similar arguments give the sharper inequality  $\gamma(E) \leq \mathcal{H}_\infty^1(E)$ .

(2) Because of Theorem 19.9, Conjecture 19.8 is now equivalent to the following.

Let  $E$  be a compact purely 1-unrectifiable subset of  $\mathbf{C}$  with  $\mathcal{H}^1(E) < \infty$ . If  $\sigma$  is a non-zero complex Radon measure on  $\mathbf{C}$  with  $\text{spt } \sigma \subset E$  and  $|\sigma(B(z, r))| \leq r$  for  $z \in \mathbf{C}$ ,  $r > 0$ , then the Cauchy transform  $C_\sigma$ ,

$$C_\sigma(z) = \int \frac{d\sigma\zeta}{\zeta - z},$$

cannot be bounded on  $\mathbf{C} \setminus E$ .

Forgetting about  $E$  we first look at some properties of  $C_\sigma$ .

**19.12. Lemma.** For any complex Radon measure  $\sigma$  on  $\mathbf{C}$ ,  $\int |\zeta - z|^{-1} d|\sigma|\zeta < \infty$  for  $\mathcal{L}^2$  almost all  $z \in \mathbf{C}$ , whence  $C_\sigma$  is defined  $\mathcal{L}^2$  almost everywhere on  $\mathbf{C}$ .

This follows immediately from Fubini's theorem and the fact that  $|z|^{-1}$  is locally integrable with respect to  $\mathcal{L}^2$ .

We introduce some notation.

**19.13. Definitions.** Let  $\sigma$  be a complex Radon measure on  $\mathbf{C}$ . For  $\varepsilon > 0$  and  $z \in \mathbf{C}$ , set

$$C_\sigma^\varepsilon(z) = \int_{\mathbf{C} \setminus B(z, \varepsilon)} \frac{d\sigma\zeta}{\zeta - z}.$$

The Cauchy maximal function of  $\sigma$  at  $z \in \mathbf{C}$  is defined by

$$C_\sigma^*(z) = \sup_{\varepsilon > 0} |C_\sigma^\varepsilon(z)|.$$

**19.14. Lemma.** Let  $\sigma$  be a complex Radon measure on  $\mathbf{C}$ ,  $M$  a positive number and  $z \in \mathbf{C}$  such that  $|\sigma|(B(z, r)) \leq Mr$  for  $r > 0$ . Then

$$|C_\sigma^*(z)| \leq \|C_\sigma\|_\infty + 20M.$$

Here  $\|C_\sigma\|_\infty$  is the usual  $L^\infty$ -norm of the almost everywhere defined function  $C_\sigma$ .

*Proof.* Suppose  $L = \|C_\sigma\|_\infty < \infty$ . For  $\varepsilon > 0$  and  $z \in \mathbf{C}$  we estimate the average

$$\begin{aligned} & \frac{1}{\pi(\varepsilon/2)^2} \int_{B(z, \varepsilon/2)} \int_{B(z, \varepsilon)} \frac{d|\sigma|\zeta}{|\zeta - w|} d\mathcal{L}^2 w \\ & \leq \int_{B(z, \varepsilon)} \frac{4}{\pi\varepsilon^2} \int_{B(\zeta, 2\varepsilon)} \frac{d\mathcal{L}^2 w}{|\zeta - w|} d|\sigma|\zeta = 16 |\sigma|(B(z, \varepsilon))/\varepsilon \leq 16M. \end{aligned}$$

Hence there is  $w \in B(z, \varepsilon/2)$  with  $|C_\sigma(w)| \leq L$  and

$$\int_{B(z, \varepsilon)} \frac{d|\sigma|\zeta}{|\zeta - w|} \leq 16M.$$

Thus

$$\begin{aligned} |C_\sigma^\varepsilon(z) - C_\sigma(w)| &= \left| \int_{\mathbf{C} \setminus B(z, \varepsilon)} \frac{d\sigma\zeta}{\zeta - z} - \int \frac{d\sigma\zeta}{\zeta - w} \right| \\ &\leq \int_{\mathbf{C} \setminus B(z, \varepsilon)} \frac{|z - w|}{|\zeta - z| |\zeta - w|} d|\sigma|\zeta + \int_{B(z, \varepsilon)} \frac{d|\sigma|\zeta}{|\zeta - w|} \\ &\leq \sum_{j=0}^{\infty} \int_{B(z, 2^{j+1}\varepsilon) \setminus B(z, 2^j\varepsilon)} \frac{|z - w|}{|\zeta - z| |\zeta - w|} d|\sigma|\zeta + 16M \\ &\leq \sum_{j=0}^{\infty} \frac{\varepsilon/2}{2^j\varepsilon(2^j\varepsilon - \varepsilon/2)} 2^{j+1}\varepsilon M + 16M \\ &\leq \left( \sum_{j=0}^{\infty} 2^{1-j} + 16 \right) M = 20M. \end{aligned}$$

This gives

$$|C_\sigma^\varepsilon(z)| \leq |C_\sigma^\varepsilon(z) - C_\sigma(w)| + |C_\sigma(w)| \leq 20M + L. \quad \square$$

## Cauchy transforms and tangent measures

We return to the situation of Theorem 19.9 with an additional assumption and look at the tangent measures (recall Chapter 14).

**19.15. Lemma.** Let  $E \subset \mathbf{C}$  be  $\mathcal{H}^1$  measurable with  $\mathcal{H}^1(E) < \infty$  and  $\Theta_*^1(E, z) > 0$  for  $\mathcal{H}^1$  almost all  $z \in E$ , and let  $\sigma$  be a complex Radon measure on  $\mathbf{C}$  with compact support satisfying  $\text{spt } \sigma \subset E$ ,  $|\sigma(B(z, r))| \leq r$  for  $z \in \mathbf{C}$ ,  $r > 0$ , and  $\|C_\sigma\|_\infty < \infty$ . Then for  $|\sigma|$  almost all  $a \in \mathbf{C}$  every  $\nu \in \text{Tan}(\mathcal{H}^1 \llcorner E, a)$  satisfies

$$\sup_{0 < \varepsilon < R < \infty} \left| \int_{B(R) \setminus B(\varepsilon)} \frac{d\nu \zeta}{\zeta} \right| < \infty.$$

*Proof.* As in the proof of Theorem 19.9,  $|\sigma|$  is absolutely continuous with respect to  $\mathcal{H}^1$  with the absolute value of the Radon–Nikodym derivative bounded by 1. Thus  $\sigma(A) = \int_A \varphi d\mathcal{H}^1$  for Borel sets  $A \subset E$  where  $\varphi: E \rightarrow \mathbf{C}$  is a Borel function with  $|\varphi| \leq 1$ .

Let  $a \in E$  with  $0 < \Theta_*^1(E, a) \leq \Theta^{*1}(E, a) < \infty$  and let  $\nu \in \text{Tan}(\mathcal{H}^1 \llcorner E, a)$ . Then there is  $M < \infty$  such that

$$(1) \quad |\sigma|(B(a, r)) \leq Mr \quad \text{for } r > 0.$$

By Remark 14.4 (3) there exist a positive number  $c$  and a sequence  $r_j \downarrow 0$  such that

$$\nu = c \lim_{j \rightarrow \infty} r_j^{-1} T_{a, r_j \#} (\mathcal{H}^1 \llcorner E).$$

The variations of the blow-ups of  $\sigma$  satisfy for  $R > 0$

$$\limsup_{j \rightarrow \infty} |r_j^{-1} T_{a, r_j \#} \sigma|(B(R)) \leq \limsup_{j \rightarrow \infty} r_j^{-1} |\sigma|(B(a, Rr_j)) \leq MR,$$

whence (see Theorem 1.23) we can extract a sub-sequence from  $(r_j^{-1} T_{a, r_j \#} \sigma)$ , which we assume to be the whole sequence, converging weakly to a complex Radon measure  $\tau$ . Using Lusin's theorem and the density theorem 6.2 (2) one sees as in the proof of Lemma 14.6 that at  $|\sigma|$  almost all points  $a \in E$ ,  $c\tau = \varphi(a)\nu$  with  $\varphi(a) \neq 0$ .

Let  $0 < \varepsilon < R < \infty$  be such that  $\nu(\partial B(\varepsilon)) = \nu(\partial B(R)) = 0$ . This holds for all but at most countably many  $\varepsilon$ 's and  $R$ 's. Then if  $\tau_i \rightarrow \tau$  weakly and  $\psi: \mathbf{C} \rightarrow \mathbf{C}$  is continuous,  $\int_{B(a, R) \setminus B(a, \varepsilon)} \psi d\tau_i \rightarrow \int_{B(a, R) \setminus B(a, \varepsilon)} \psi d\tau$  as one easily verifies. Thus we compute

$$\begin{aligned} \left| c^{-1} \varphi(a) \int_{B(R) \setminus B(\varepsilon)} \frac{d\nu \zeta}{\zeta} \right| &= \left| \int_{B(R) \setminus B(\varepsilon)} \frac{d\tau \zeta}{\zeta} \right| \\ &= \lim_{j \rightarrow \infty} r_j^{-1} \left| \int_{B(R) \setminus B(\varepsilon)} \frac{dT_{a, r_j \#} \sigma}{\zeta} \right| \\ &= \lim_{j \rightarrow \infty} \left| \int_{B(a, r_j R) \setminus B(a, r_j \varepsilon)} \frac{d\sigma \zeta}{\zeta - a} \right| \\ &\leq \limsup_{j \rightarrow \infty} 2 |C_\sigma^*(a)| \leq 2 \|C_\sigma\|_\infty + 40M, \end{aligned}$$

where the last inequality follows from Lemma 19.14. By approximation the same estimate holds for all  $0 < \varepsilon < R < \infty$ , which proves the lemma.  $\square$

**19.16. Remark.** If we assume the stronger condition  $|\sigma|(B(z, r)) \leq r$  for  $z \in \mathbf{C}$ ,  $r > 0$ , a similar proof gives that for  $|\sigma|$  almost all  $a \in \mathbf{C}$  every  $\nu \in \text{Tan}(\mathcal{H}^1 \llcorner E, a)$  satisfies

$$\sup_{0 < \varepsilon < R < \infty} \left| \int_{B(z, R) \setminus B(z, \varepsilon)} \frac{d\nu \zeta}{\zeta - z} \right| < \infty \quad \text{for } z \in \mathbf{C}.$$

### Analytic capacity and rectifiability

We can now prove a partial result related to Conjecture 19.8.

**19.17. Theorem.** Let  $E$  be a compact subset of  $\mathbf{C}$  with  $\mathcal{H}^1(E) < \infty$ . If for  $\mathcal{H}^1$  almost all  $a \in E$ ,  $\Theta_*^1(E, a) > 0$  and for every  $\nu \in \text{Tan}(\mathcal{H}^1 \llcorner E, a)$  the support of  $\nu$  is not contained in a line, then  $\gamma(E) = 0$ .

*Remark.* By Theorem 16.5  $E$  is purely 1-unrectifiable. It is not difficult to show that the above condition on the tangent measures is equivalent to the following: for  $\mathcal{H}^1$  almost all  $a \in E$ ,  $\text{Tan}(\mathcal{H}^1 \llcorner E, a) \cap \mathcal{G}_{2,1} = \emptyset$ .

*Proof of Theorem.* Our assumptions imply the following: for  $\mathcal{H}^1$  almost all  $a \in E$  and for all  $\nu \in \text{Tan}(\mathcal{H}^1 \llcorner E, a)$ , there is  $s$ ,  $0 < s < 1/2$ , such that for all  $r > 0$  and  $L \in G(2, 1)$

$$(1) \quad \text{spt } \nu \cap B(r) \setminus X(0, L, s) \neq \emptyset.$$

Recall 15.12 for the notation. Otherwise a suitable tangent measure of  $\nu$  at 0, which would also be a tangent measure of  $\mathcal{H}^1 \llcorner E$  at  $a$  by Theorem 14.16, would have its support contained in a line. Note that  $0 \in \text{spt } \nu$  and  $\nu$  fulfils the uniform density estimates of Lemma 14.7. In particular, at  $\mathcal{H}^1$  almost all points  $a$  of  $E$  every  $\nu \in \text{Tan}(\mathcal{H}^1 \llcorner E, a)$  satisfies

$$(2) \quad \nu(B(z, r)) \geq cr \quad \text{for } z \in \text{spt } \nu, \quad r > 0,$$

where  $c$  is a positive number depending on  $a$  and  $\nu$ .

Suppose  $\gamma(E) > 0$ . By the definition of  $\gamma(E)$  and Theorem 19.9 we can find a non-zero complex Radon measure  $\sigma$  such that  $\text{spt } \sigma \subset E$ ,

$|\sigma(B(z, r))| \leq r$  for  $z \in \mathbf{C}$ ,  $r > 0$ , and  $\|C_\sigma\|_\infty < \infty$ . By Lemma 19.15 at  $|\sigma|$  almost all points  $a \in E$  every  $\nu \in \text{Tan}(\mathcal{H}^1 \llcorner E, a)$  satisfies

$$(3) \quad \sup_{\varepsilon > 0} \left| \int_{B(1) \setminus B(\varepsilon)} \zeta^{-1} d\nu \zeta \right| < \infty.$$

Applying Lemma 14.7 we can finally find  $a \in E$  and  $\nu \in \text{Tan}(\mathcal{H}^1 \llcorner E, a)$  such that the above conditions (1)–(3) hold and  $\text{spt } \nu$  is contained in some half-plane with 0 as a boundary point. Rotating  $E$ , we may assume

$$(4) \quad \text{spt } \nu \subset \{x + iy : y \leq 0\}.$$

By (1), there are  $s$ ,  $0 < s < 1/2$ , and points  $z_j \in B(2^{-j}) \cap \text{spt } \nu$  such that  $|z_1| \geq |z_2| \geq \dots$ , the discs  $B_j = B(z_j, s|z_j|/2)$  are disjoint and contained in  $\{z : -\text{Im } z \geq s|z|/2\}$ , where  $\text{Im } z$  denotes the imaginary part of  $z$ . Then (4) and (2) give for  $k = 1, 2, \dots$ ,

$$\begin{aligned} \left| \int_{B(1) \setminus B(|z_k|)} \zeta^{-1} d\nu \zeta \right| &\geq \int_{B(1) \setminus B(|z_k|)} \frac{\text{Im } \bar{\zeta}}{|\zeta|^2} d\nu \zeta \\ &\geq \sum_{j=1}^{k-1} \int_{B_j} (-\text{Im } \zeta / |\zeta|^2) d\nu \zeta \geq \frac{s}{2} \sum_{j=1}^{k-1} \int_{B_j} |\zeta|^{-1} d\nu \zeta \\ &\geq \frac{s}{2} \sum_{j=1}^{k-1} \frac{\nu(B_j)}{2|z_j|} \geq s^2 c(k-1)/8. \end{aligned}$$

Letting  $k \rightarrow \infty$ , we obtain a contradiction to (3).  $\square$

### Various remarks

**19.18.** (1) There are many purely 1-unrectifiable sets which satisfy the assumptions of Theorem 19.17. For example, if  $E$  is a one-dimensional self-similar set and  $E$  is not a line segment this is easily verified, or see Mattila [5].

(2) The tangent measure condition of Theorem 19.17 can be somewhat relaxed. One can use instead of Lemma 14.7(3) the following lower density theorem in half-planes, due to Besicovitch [3]: if  $A \subset \mathbf{R}^2$  is  $\mathcal{H}^1$  measurable, purely 1-unrectifiable,  $\mathcal{H}^1(A) < \infty$  and  $e \in S^1$ , then  $\Theta_*^1(\mathcal{H}^1 \llcorner \{x \in A : (x - a) \cdot e \geq 0\}, a) = 0$  for  $\mathcal{H}^1$  almost all  $a \in A$ . Then a similar proof to that of Theorem 19.17 gives the following.

Let  $E$  be a compact purely 1-unrectifiable subset of  $\mathbf{C}$  with  $\mathcal{H}^1(E) < \infty$ . Suppose that for  $\mathcal{H}^1$  almost all  $a \in E$ ,  $\Theta_*^1(E, a) > 0$  and there

is  $L \in G(2, 1)$ , depending on  $a$ , such that the support of no tangent measure of  $\mathcal{H}^1 \llcorner E$  at  $a$  is contained in  $L$ . Then  $\gamma(E) = 0$ .

For details, see Mattila [8]. There are examples of purely 1-unrectifiable compact sets  $E$  with  $\mathcal{H}^1(E) < \infty$  for which  $\Theta_*^1(E, a) > 0$  for all  $a \in E$  and the above condition is not satisfied. That is,  $\mathcal{G}_{2,1} \subset \text{Tan}(\mathcal{H}^1 \llcorner E, a)$  for  $\mathcal{H}^1$  almost all  $a \in E$ . One is given by the construction of Example 14.2 (3). For these particular circle sets Fang [1] showed by a different method that  $\gamma(E) = 0$ . For some others, see Fang [1]–[2].

(3) The first example of a compact set  $E$  with  $\mathcal{H}^1(E) > 0$  and  $\gamma(E) = 0$  was given by Vitushkin [1]. Later on, Garnett [1], and independently L. D. Ivanov, proved that  $\gamma(E) = 0$  for  $E = C(1/4) \times C(1/4)$  where  $C(1/4)$  is as in 4.10. Other proofs for this set have been given by Murai [1] and Jones [2], see also Christ [1]. The above arguments are based on Mattila [8]. There are also many sets  $E$  with non- $\sigma$ -finite  $\mathcal{H}^1$  measure and zero analytic capacity. Garnett [2, § IV.2] attempted to characterize decreasing sequences  $T = (\lambda_i)$  for which  $E = C(T) \times C(T)$ , recall 4.11, has zero analytic capacity. There is a gap in the argument, as was observed by V. Eiderman, but still Garnett's condition  $C_1(E) = 0$  could be equivalent with  $\gamma(E) = 0$  in this case.

(4) If one believes in Conjecture 19.8 one might hope that some of the equivalent conditions for a set  $E$  with  $\mathcal{H}^1(E) < \infty$  to be purely 1-unrectifiable would characterize zero analytic capacity also for compact sets  $E$  of non- $\sigma$ -finite  $\mathcal{H}^1$  measure. But one notices very quickly that among these only the projection condition,  $\mathcal{H}^1(P_L E) = 0$  for  $\gamma_{2,1}$  almost all  $L \in G(2, 1)$ , that is,  $\mathcal{I}_1^1(E) = 0$ , might do that. Vitushkin [2] conjectured this, but it also turned out to be false. In Mattila [11] it was shown that this condition is not conformally invariant (recall 5.14) so it cannot be the right one, but this does not say which of the two possible implications is false. Later Jones and Murai [1] gave an example of  $E$  with  $\gamma(E) > 0$  and  $\mathcal{I}_1^1(E) = 0$ . It is still not known whether  $\gamma(E) = 0$  implies  $\mathcal{I}_1^1(E) = 0$ . A deep analysis of the relations between projection properties, analytic capacity and the behaviour of Cauchy transform can be found in the book of Murai [2].

(5) The positiveness of analytic capacity is very closely related to the boundedness of Cauchy integral as an operator on  $L^2$ . This was the last key step required to complete the proof of Theorem 19.6. The relation can be explained more generally. Let  $E \subset \mathbb{C}$  be compact and assume that for some  $0 < c < d < \infty$ ,

$$(a) \quad cr \leq \mathcal{H}^1(E \cap B(z, r)) \leq dr \quad \text{for } z \in E, \ 0 < r \leq 1.$$

Formally the Cauchy integral operator  $C_E$  on  $E$  is  $C_E f(z) = \int_E (\zeta - z)^{-1} f(\zeta) d\mathcal{H}^1 \zeta$ , but of course the integral is not usually defined at the points of  $E$ . By saying that this operator is bounded on  $L^2$  we mean that there is  $M < \infty$  such that

$$\sup_{\varepsilon > 0} \int_E \left| \int_{E \setminus B(z, \varepsilon)} (\zeta - z)^{-1} f(\zeta) d\mathcal{H}^1 \zeta \right|^2 d\mathcal{H}^1 z \leq M \int_E |f|^2 d\mathcal{H}^1$$

for all  $f \in L^2(\mathcal{H}^1 \llcorner E)$ .

Assuming (a) the following results hold, see Christ [1]–[2].

If  $C_E$  is bounded on  $L^2$ , then  $\gamma(E) > 0$ .

Conversely, if  $\gamma(E) > 0$ ,  $E$  has a compact subset  $F$  with  $\mathcal{H}^1(F) > 0$  for which  $C_F$  is bounded on  $L^2$ .

The following question is open: if (a) holds and  $C_E$  is bounded on  $L^2$ , must  $E$  be 1-rectifiable (or perhaps even uniformly 1-rectifiable, recall 15.23)? According to the above remarks this is a weaker form of Conjecture 19.8. If one assumes additionally that  $E$  satisfies condition (b) of 16.8 (3), then  $E$  is uniformly 1-rectifiable, see David and Semmes [2, Theorem I.2.33], and Fang [1]–[2] for a related result. If the Cauchy kernel  $z^{-1}$  is replaced by  $x|x|^{-m-1}$ ,  $x \in \mathbf{R}^n$ , the corresponding result holds for  $m$ -dimensional sets in  $\mathbf{R}^n$ .

We shall discuss more relations between singular integrals and rectifiability in the next chapter. Here we state a characterization for uniform 1-rectifiability as a quadratic estimate for the Cauchy transform. This corresponds to square function estimates which are central in various areas of analysis, see e.g. Stein [1] and Jones [2].

Let  $E$  be as in 15.23 (1) with  $m = 1$ . Then  $E$  is uniformly 1-rectifiable if and only if there exists  $C < \infty$  such that

$$\int_{\mathbf{C}} |F'(z)|^2 d(z, E) d\mathcal{L}^2 z \leq C \int_E |f|^2 d\mathcal{H}^1$$

whenever  $f \in L^2(\mathcal{H}^2 \llcorner E)$  and

$$F(z) = \int_E (\zeta - z)^{-1} f(\zeta) d\mathcal{H}^1 \zeta \quad \text{for } z \in \mathbf{C} \setminus E.$$

An analogous result holds for  $(n-1)$ -dimensional sets if the kernel  $z^{-1}$  is replaced by  $x|x|^{1-n}$ , see David and Semmes [2, Theorem I.2.41]. The proof is based on the convexity characterization mentioned in 16.8 (3).

(6) A very similar problem in  $\mathbf{R}^n$  seems to be the removability for harmonic Lipschitz functions. We say that a compact subset  $E$  of  $\mathbf{R}^n$  is removable for harmonic Lipschitz functions, abbreviated RHL, if whenever  $U$  is an open set containing  $E$  every Lipschitz function  $u: U \rightarrow \mathbf{R}$  which is harmonic in  $U \setminus E$  must be harmonic in  $U$ .

In  $\mathbf{R}^2 = \mathbf{C}$  the complex derivative  $\partial_z u$  of a harmonic Lipschitz function  $u$  is a bounded analytic function. Hence  $\gamma(E) = 0$  implies that  $E$  is RHL. It is not known whether the converse holds.

Again it is easy to show that for a compact set  $E \subset \mathbf{R}^n$ ,  $n \geq 2$ ,  $\mathcal{H}^{n-1}(E) = 0 \implies E$  is RHL  $\implies \dim E \leq n - 1$ . Uy [4] showed that in  $\mathbf{R}^n$ , for any  $n \geq 2$ , there are compact RHL-sets with  $\mathcal{H}^{n-1}(E) > 0$ . Other examples were given in Mattila and Paramonov [1]. Analogues of Theorems 19.6 and 19.17 were established for RHL-sets by Uy [1], [3] and Mattila and Paramonov [1]. For a capacity, analogous to the analytic capacity and related to RHL-sets, see Paramonov [1].

(7) Besicovitch [3] showed that if  $U$  is an open subset of  $\mathbf{C}$  and the set of points  $E_f$  where a bounded function  $f: U \rightarrow \mathbf{C}$  fails to have a complex derivative has zero  $\mathcal{H}^1$  measure, then  $f$  agrees in  $U \setminus E_f$  with a function which is analytic in  $U$ . The set  $E_f$  need not be closed; in fact, it can be dense in  $U$ . He also showed that if  $f$  is continuous in  $U$  and  $E_f$  has  $\sigma$ -finite  $\mathcal{H}^1$  measure, then  $f$  is analytic in  $U$ . See also Howard [1].

Kaufman [6] extended Besicovitch's results from bounded and continuous functions to functions in BMO and VMO, respectively. Moreover, he showed that in these classes the above conditions on the  $\mathcal{H}^1$  measure of  $E_f$  are also necessary.

(8) Khavinson has found interesting connections between geometric measure theory, Cauchy transforms and rational approximation. For instance, he has discovered a proof for the isoperimetric inequality in the plane via this route, see Khavinson [1]–[2] and Gamelin and Khavinson [1].

## Exercises.

1. Show that  $r/2 \leq \gamma(B(z, r)) \leq 2r$  for  $z \in \mathbf{C}$ ,  $0 < r < \infty$ .
2. Let  $E = C(1/4) \times C(1/4)$  with  $C(1/4)$  as in 4.10, and let  $\mu = \mathcal{H}^1 \llcorner E$ . Show directly, without using the general results, that  $C_\mu$  is not bounded on  $\mathbf{C} \setminus E$ .
3. Let  $\mu = \mathcal{H}^1 \llcorner S^1$ . Is  $C_\mu$  bounded on  $\mathbf{C} \setminus B(1)$ ?
4. Let  $s > 1$  and let  $\mu$  be a Radon measure on  $\mathbf{C}$  with compact support such that  $\mu(B(z, r)) \leq r^s$  for  $z \in \mathbf{C}$  and  $r > 0$ . Show

that  $C_\mu$  is a bounded non-constant analytic function in  $\mathbf{C} \setminus \text{spt } \mu$  and that it is also Hölder continuous with exponent  $s - 1$ .

5. Do the computation for the proof of Theorem 19.6 in the case  $\Gamma \subset \mathbf{R}$ .